

Master/
Orchestrator

SU#

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IJCSA: Solve Algebraic Loop at time t

Initialize $U^{(0)}$ from current inputs

loop

[until convergence]

1. Evaluate subsystems in parallel

par

[Parallel Execution]

evaluate_outputs($U1^{(m)}, t$)

$Y1^{(m+1)} = S1(U1^{(m)})$

Compute with $dt=0$
(frozen state)

$Y1^{(m+1)}$

evaluate_outputs($U2^{(m)}, t$)

$Y2^{(m+1)} = S2(U2^{(m)})$

Compute with $dt=0$

$Y2^{(m+1)}$

2. Compute residual

$r^{(m)} = [U1^{(m)} - Y2^{(m+1)}$
 $[U2^{(m)} - Y1^{(m+1)}]$

3. Check convergence

alt

[$\|r^{(m)}\| < \text{tolerance}$]

Converged

Exit iteration

4. Compute interface Jacobian

$J^{(m)} = dR/dU$ via finite differences

For each interface input:
perturb U_j , evaluate outputs,
compute $J[:, j] = (R_{\text{pert}} - R) / \text{eps}$

5. Solve for correction

$J^{(m)} \cdot \#U^{(m)} = -r^{(m)}$

6. Apply correction

$U^{(m+1)} = U^{(m)} + \#U^{(m)}$

7. Commit converged solution

set_inputs($U1^*, t$)

inputs committed

set_inputs($U2^*, t$)

inputs committed

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